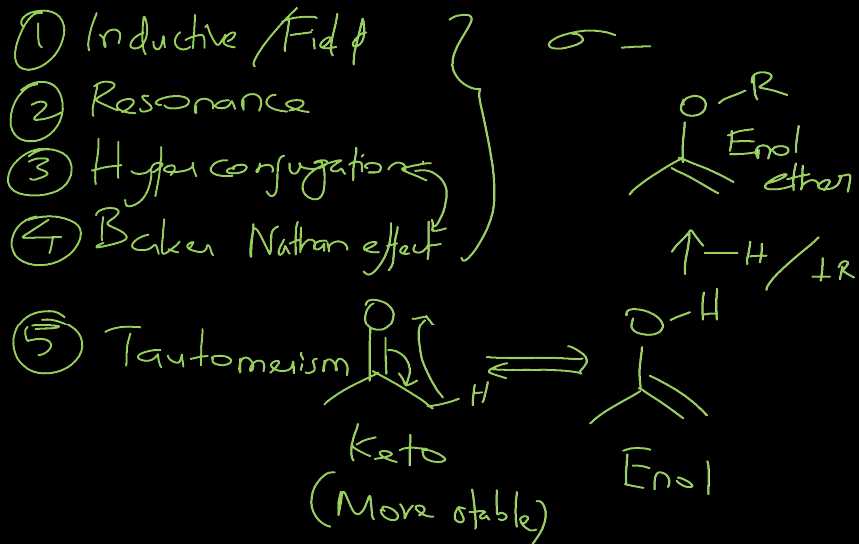
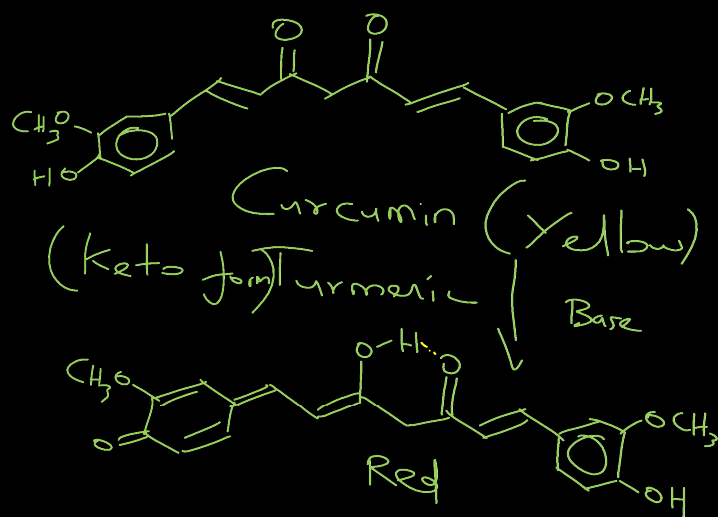
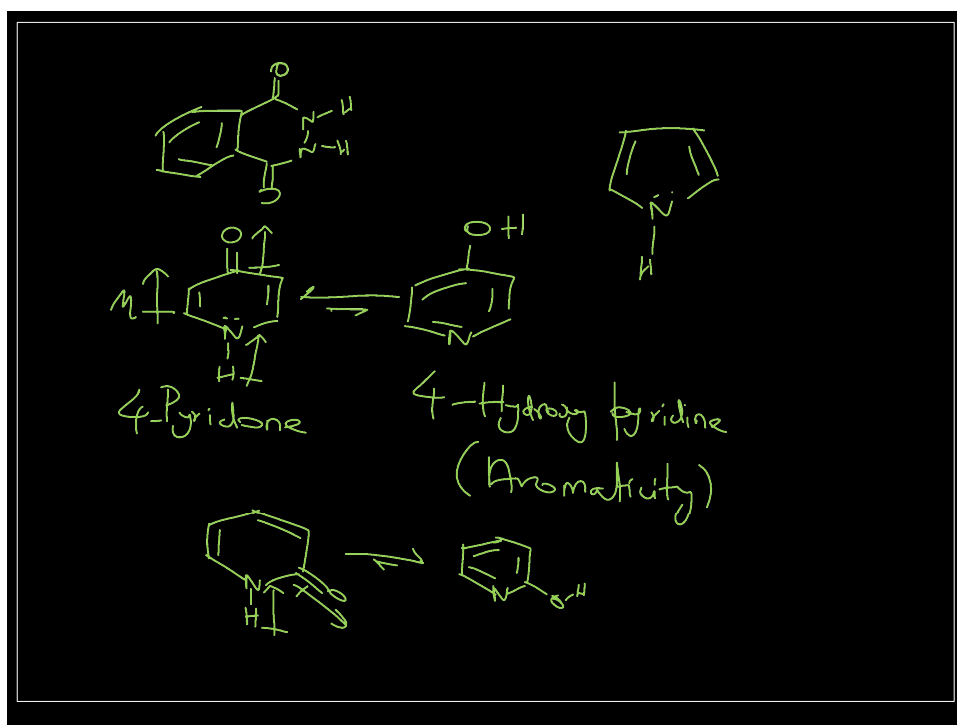
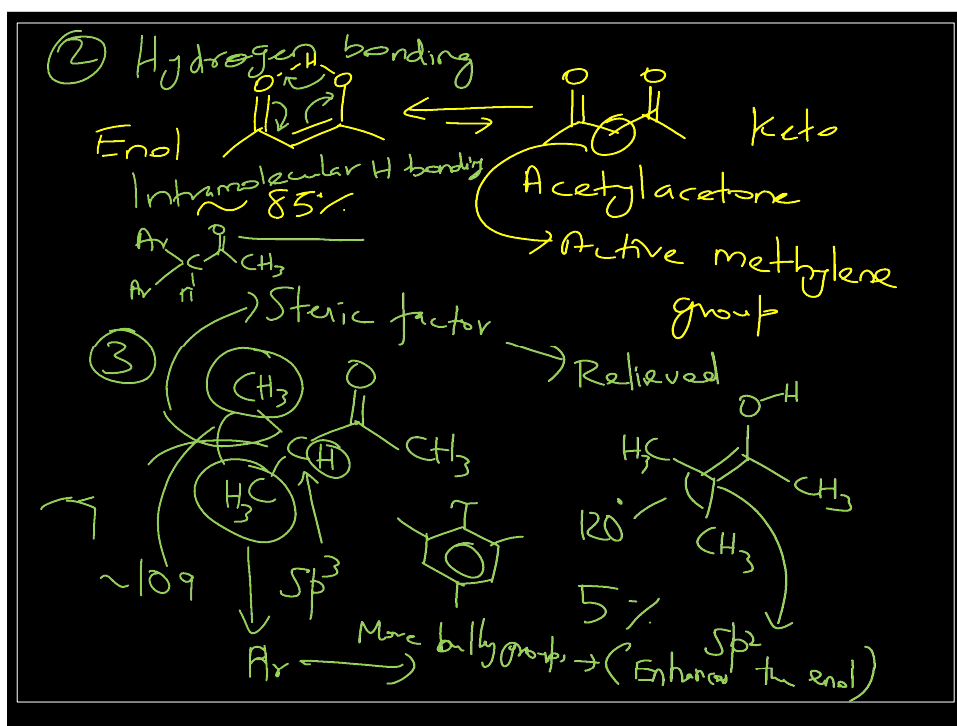
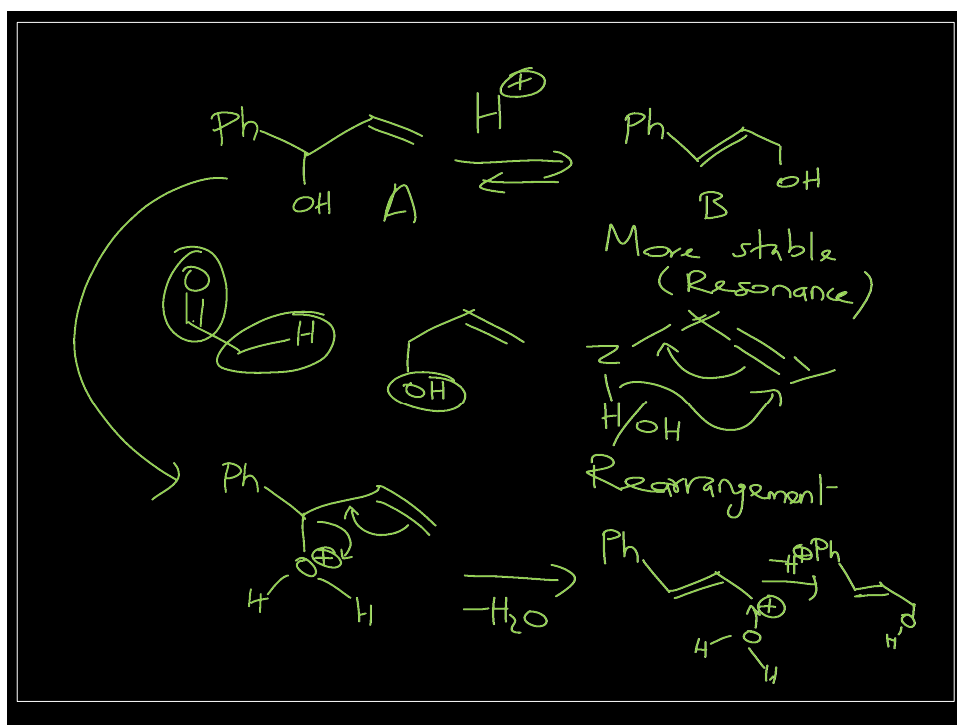
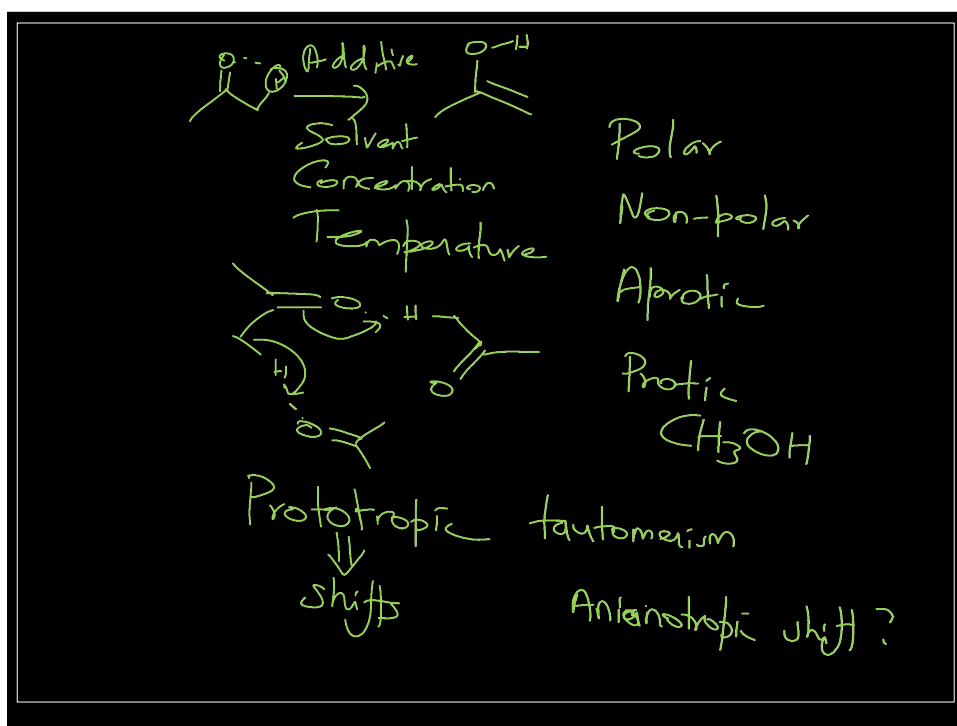


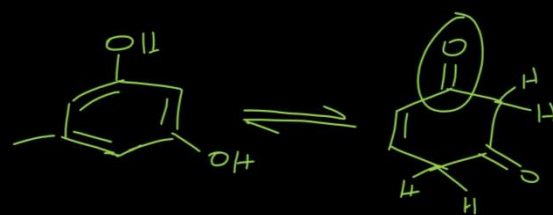
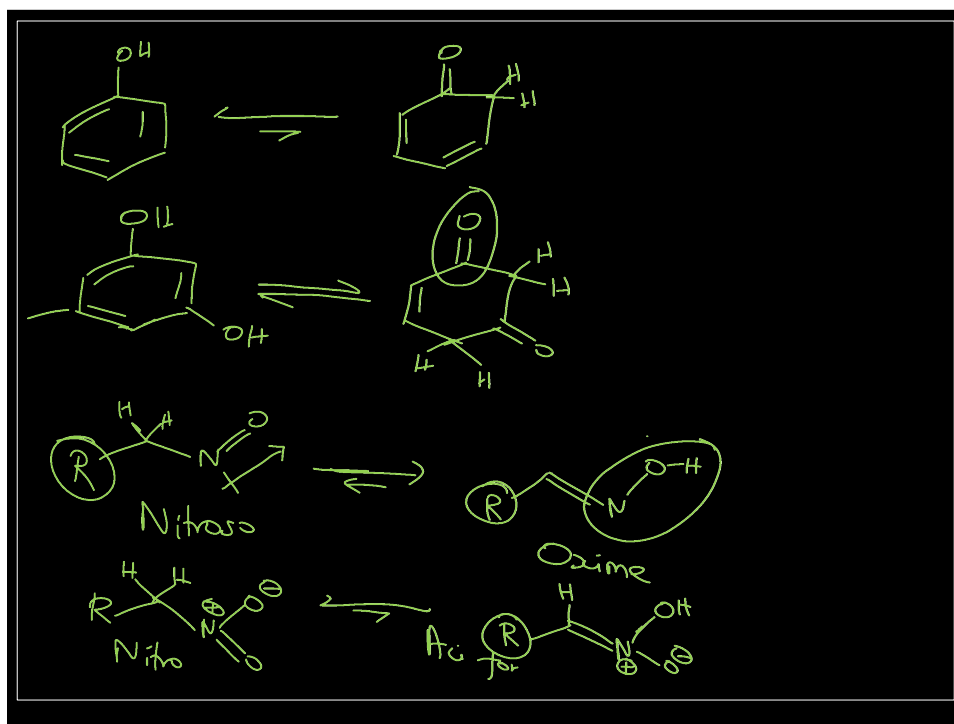


① Acid or base catalysed









Simple calculation leads to a ΔH^0 value close to zero, essentially because two strongly bonded carbonyl groups are present in the keto structure to offset the aromatic stabilization energy of the enol. Consistent with this, *m*-dihydroxybenzene has properties characteristic of both a phenol and a ketone: for example, it undergoes the rapid electrophilic substitutions such as bromination which are characteristic of phenols (p. 386) and it is reduced by sodium amalgam to 1,3-cyclohexanedione in the manner characteristic of $\alpha\beta$ -unsaturated carbonyl compounds (p. 638). An extension of the argument rationalizes the behaviour of 1,3,5-trihydroxybenzene, which is more fully ketonized.

